

AP Statistics FRQ Practice Exam 05

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AP Statistics FRQ Practice Exam 5

APSTAT-FRQ5-001

Section: FRQ

Type: frq

Primary Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Secondary Unit: Unit 3, Inference for Categorical Data: Proportions

FRQ Type: Formulate Questions and Collect Data

Prompt:

Researchers recruit 90 adult volunteers from one community who report that they frequently experience nausea in virtual-reality environments. The researchers want to study whether an active acupressure wristband helps users complete a standardized 20-minute virtual-reality ride without ending early because of nausea.

Each volunteer receives either an active wristband or a visually identical sham wristband. The wristbands are supplied in coded packages. The participants and the evaluator who records whether each ride is completed do not know which type was used. Results are shown.

Treatment	Completed ride	Ended early	Total
Active wristband	34	11	45
Sham wristband	24	21	45

- (a) Identify the population of interest, the sample, and the experimental units.
- (b) Identify the treatments and response variable. State a statistical question the experiment is designed to answer.
- (c) Describe a valid random-assignment procedure for placing the 90 volunteers into two groups of 45. Explain the purpose of random assignment.
- (d) Explain the purposes of the sham wristband and the blinding used in the experiment.
- (e) Calculate and interpret the observed difference in completion proportions, active minus sham. State the causal and generalization conclusions that this design could support if the difference is judged meaningful.

Scoring Guide:

Total Points: 10

(a) 2 points

- Point 1: Identifies the population of interest as adults who frequently experience nausea in virtual-reality environments and the sample as the 90 recruited volunteers from the community.
- Point 2: Identifies the 90 individual volunteers as the experimental units.

(b) 2 points

- Point 1: Identifies the active and sham wristbands as the treatments and ride completion without ending early because of nausea as the binary response variable.
- Point 2: States a comparative statistical question about whether the probability or proportion completing the ride differs between active- and sham-wristband conditions.

(c) 2 points

- Point 1: Describes a valid chance process, such as randomly shuffling all 90 participant labels and assigning the first 45 to active and the remaining 45 to sham.
- Point 2: Explains that random assignment tends to balance preexisting characteristics and other potential confounders between groups, supporting a cause-and-effect comparison.

(d) 2 points

- Point 1: Explains that the visually identical sham controls for placebo effects and for the experience of wearing a wristband.
- Point 2: Explains that blinding participants limits response or expectation effects and blinding the evaluator limits differential outcome assessment.

(e) 2 points

- Point 1: Calculates $\hat{p}_{\text{active}} - \hat{p}_{\text{sham}} = 34/45 - 24/45 = 10/45 \approx 0.222$ and interprets it as a completion rate about 22.2 percentage points higher in the active group.
- Point 2: States that random assignment can support a causal conclusion for volunteers like those studied, but volunteer recruitment does not support generalization to all adults who experience virtual-reality nausea.

Model Guidance:

The volunteers are both the sampled people and the experimental units, while the broader target is adults prone to virtual-reality nausea. Randomly assign participant labels to two equal groups before distributing coded bands. This creates treatment groups that should be similar on average with respect to susceptibility, prior experience, and other background variables.

The sham band gives both groups the same visible treatment experience, and the two forms of blinding reduce expectation and evaluator effects. The observed completion proportions are $34/45 \approx 0.756$ and $24/45 \approx 0.533$, a difference of about 0.222. The experiment can attribute a meaningful difference to the active band among the volunteers because treatment was randomized. Since the participants volunteered rather than being randomly sampled from the target population, broader generalization requires caution.

Expected Points:

10

APSTAT-FRQ5-002

Section: FRQ

Type: frq

Primary Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Secondary Unit: Unit 1, Exploring One-Variable Data and Collecting Data

FRQ Type: Analyze Data, One- or Two-Variable Data and Probability

Prompt:

A sports-science center records choice-reaction times for two standardized goalkeeper drills. Reaction times for athletes in the center's reference population are each adequately modeled by a normal distribution. Lower times indicate faster reactions.

Drill	Mean	Standard deviation	One goalkeeper's time
Visual-choice drill	480 ms	40 ms	420 ms
Auditory-choice drill	360 ms	30 ms	318 ms

- (a) Calculate the goalkeeper's z-score for each drill. Determine in which drill the goalkeeper performed better relative to the corresponding reference distribution.
- (b) Calculate the percentile of the goalkeeper's reaction time in each drill. Interpret the visual-choice percentile in context, accounting for the fact that lower times are better.
- (c) For a randomly selected athlete from the visual-choice reference population, calculate the probability of a reaction time between 440 ms and 520 ms. Interpret the result.
- (d) The center defines the fastest 5% on the visual-choice drill as elite. The z-score with 0.05 area to its left is -1.645 . Determine the reaction-time cutoff for elite status and whether the goalkeeper qualifies.
- (e) Suppose a later histogram of visual-choice times is strongly right-skewed because a few athletes have very slow responses. Explain how that finding would affect the normal-model calculations in parts (b) through (d).

Scoring Guide:

Total Points: 10

(a) 2 points

- Point 1: Calculates $z_{\text{visual}} = (420 - 480)/40 = -1.50$ and $z_{\text{auditory}} = (318 - 360)/30 = -1.40$.
- Point 2: Concludes that the visual-choice performance is better relative to its reference group because -1.50 represents a reaction time farther below the mean and lower times are better.

(b) 2 points

- Point 1: Calculates $P(Z \leq -1.50) \approx 0.0668$ and $P(Z \leq -1.40) \approx 0.0808$, placing the times at about the 6.7th and 8.1st percentiles, respectively.
- Point 2: Interprets the visual result: about 6.7% of reference athletes have times at or below 420 ms, so this goalkeeper is faster than about 93.3% of that reference population.

(c) 2 points

- Point 1: Standardizes 440 ms and 520 ms as $z = -1$ and $z = 1$ and calculates $P(-1 \leq Z \leq 1) \approx 0.6827$.
- Point 2: Interprets the result: about 68.3% of athletes in the visual-choice reference population are expected to have reaction times from 440 ms through 520 ms.

(d) 2 points

- Point 1: Calculates the cutoff as $x = 480 + (-1.645)(40) = 414.2$ ms.
- Point 2: Concludes that elite status requires a time at or below about 414.2 ms, so the goalkeeper's 420 ms does not qualify.

(e) 2 points

- Point 1: Explains that strong right skew conflicts with the stated normal model, especially for percentile and tail calculations.
- Point 2: States that the numerical probabilities and 5% cutoff could then be inaccurate and that the center should use the empirical distribution or another suitable model instead.

Model Guidance:

Standardization allows comparison across drills with different means and standard deviations. The goalkeeper is 1.50 standard deviations faster than the visual mean and 1.40 standard deviations faster than the auditory mean, so the visual result is relatively stronger. Normal-table areas to the left are approximately 0.0668 and 0.0808. Because short times are desirable, a low percentile corresponds to strong performance.

For the visual drill, 440 ms and 520 ms are one standard deviation below and above the mean, giving a central probability near 0.6827. The fastest 5% lie below $480 - 1.645(40) = 414.2$ ms; 420 ms is not fast enough for that cutoff. These results depend on a credible normal model. Strong right skew would make normal tail areas and a normal-based elite cutoff questionable.

Expected Points:

10

APSTAT-FRQ5-003

Section: FRQ

Type: frq

Primary Unit: Unit 4, Inference for Quantitative Data: Means

Secondary Unit: Unit 1, Exploring One-Variable Data and Collecting Data

FRQ Type: Analyze Data, Statistical Inference (Confidence Interval or Significance Test)

Prompt:

A meal-delivery company buys insulated lunch containers from two suppliers. For each supplier, a quality analyst takes an independent simple random sample from that supplier's current production. The response is the number of hours a sealed container keeps its contents above a specified food-safe temperature in a standardized laboratory test.

Supplier	Sample size, n	Sample mean, $\bar{x}$	Sample standard deviation, s
A	22	9.84 hours	0.62 hour
B	20	9.31 hours	0.71 hour

Side-by-side dotplots for the two samples are roughly symmetric with no apparent outliers. Each supplier produced more than 220 containers during the sampling period. For a 90% confidence interval using the conservative degrees of freedom $df = 38$, use $t^* = 1.686$.

- (a) Define the population parameter to be estimated and identify the appropriate confidence-interval procedure.
- (b) Check the conditions for the confidence-interval procedure.
- (c) Calculate the standard error and margin of error, and construct a 90% confidence interval for $\mu_A - \mu_B$.
- (d) Interpret the confidence interval in context.
- (e) A purchasing manager claims that Supplier A's containers have a mean safe-temperature duration at least 0.25 hour longer than Supplier B's. Use the interval to evaluate the claim.

Scoring Guide:

Total Points: 10

(a) 2 points

- Point 1: Defines $\mu_A - \mu_B$ as the difference between the true mean safe-temperature durations for the current production of containers from Suppliers A and B.
- Point 2: Selects an independent two-sample t confidence interval for a difference in population means.

(b) 3 points

- Point 1: Cites the two independent simple random samples and the independence of the supplier groups.
- Point 2: Verifies the 10% condition because 22 and 20 are no more than 10% of the corresponding suppliers' production totals.
- Point 3: Uses the roughly symmetric, outlier-free dotplots to justify t procedures for the two relatively small samples.

(c) 3 points

- Point 1: Calculates $SE = \sqrt{[(0.62^2/22) + (0.71^2/20)]} \approx 0.2066$ hour.
- Point 2: Calculates the margin of error as $1.686(0.2066) \approx 0.3483$ hour.
- Point 3: Constructs $(9.84 - 9.31) \pm 0.3483$, giving the interval (0.182, 0.878) hour, allowing for minor rounding differences.

(d) 1 point

- Point 1: States that the analyst is 90% confident that Supplier A's current-production containers have a true mean safe-temperature duration between about 0.182 and 0.878 hour longer than Supplier B's.

(e) 1 point

- Point 1: States that the interval supports a positive mean advantage for Supplier A but does not support an advantage of at least 0.25 hour because plausible differences from approximately 0.182 to 0.878 are included.

Model Guidance:

The samples come independently from two production populations, and the response is quantitative with unknown population standard deviations. Therefore, use a two-sample t interval for $\mu_A - \mu_B$. Random sampling, the 10% checks, and the roughly symmetric sample distributions without outliers support the method.

The observed difference is 0.53 hour. Its estimated standard error is about 0.2066, and multiplication by $t^* = 1.686$ gives a margin of error near 0.3483. The 90% interval is approximately (0.182, 0.878) hour. All values are positive, so Supplier A likely has the larger population mean for current production. However, the interval contains positive differences smaller than 0.25 hour, so it does not establish the manager's stronger minimum-advantage claim.

Expected Points:

10

APSTAT-FRQ5-004

Section: FRQ

Type: frq

Primary Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Secondary Unit: Unit 2, Probability, Random Variables, and Probability Distributions

FRQ Type: Interpret Results, Integrated Analysis and Justification

Prompt:

A small theater has five stage backdrops. The hours required to repaint the five backdrops during the previous season were

6, 8, 10, 13, 18.

The theater manager will select a simple random sample of 2 different backdrops, without replacement, and use the sample mean $\bar{x}$ to estimate the mean repainting time for all 5 backdrops. There are $C(5, 2) = 10$ equally likely unordered samples.

(a) Calculate the population mean and median repainting times. Compare the two measures of center in light of the data pattern.

(b) List all 10 possible samples of 2 backdrops and calculate $\bar{x}$ for each. Construct the probability distribution of $\bar{x}$.

(c) On paper, sketch a dotplot of the sampling distribution of $\bar{x}$. Label the axis, and state the center and range of the sampling distribution.

(d) Calculate the mean of the sampling distribution of $\bar{x}$. Explain what the result shows about $\bar{x}$ as an estimator of the population mean.

(e) Calculate the probability that $\bar{x}$ is within 2 hours, inclusive, of the population mean. Use the probability to evaluate the manager's claim that an SRS of 2 has at least a 75% chance of estimating the population mean within 2 hours.

Scoring Guide:

Total Points: 10

(a) 2 points

• Point 1: Calculates the population mean $\mu = (6 + 8 + 10 + 13 + 18)/5 = 11$ hours and the median as 10 hours.

· Point 2: Explains that the high value 18 pulls the mean above the median, consistent with a distribution having a longer upper side.

(b) 2 points

· Point 1: Lists the 10 samples and means correctly: (6, 8) → 7; (6, 10) → 8; (6, 13) → 9.5; (6, 18) → 12; (8, 10) → 9; (8, 13) → 10.5; (8, 18) → 13; (10, 13) → 11.5; (10, 18) → 14; (13, 18) → 15.5.

· Point 2: Gives the sampling distribution with $\bar{x}$ values 7, 8, 9, 9.5, 10.5, 11.5, 12, 13, 14, and 15.5, each having probability 0.10.

(c) 2 points

· Point 1: Draws one dot at each of the 10 possible $\bar{x}$ values on a horizontal axis labeled “sample mean repainting time, $\bar{x}$ (hours).”

· Point 2: States that the sampling distribution is centered at 11 hours and ranges from 7 to 15.5 hours, for a range of 8.5 hours.

(d) 2 points

· Point 1: Calculates $\mu\bar{x} = [7 + 8 + 9 + 9.5 + 10.5 + 11.5 + 12 + 13 + 14 + 15.5]/10 = 11$ hours.

· Point 2: Explains that the sampling-distribution mean equals the population mean, showing that $\bar{x}$ is an unbiased estimator of μ under this SRS process.

(e) 2 points

· Point 1: Identifies 6 qualifying means from 9 through 13 inclusive and calculates $P(|\bar{x} - 11| \leq 2) = 6/10 = 0.60$.

· Point 2: Rejects the manager's claim because 0.60 is less than 0.75; the exact sampling distribution gives only a 60% chance of being within 2 hours.

Model Guidance:

The population mean is 11 hours and the median is 10 hours. The 18-hour backdrop raises the mean above the median. For an SRS of two, enumerate unordered pairs rather than ordered selections; each of the 10 pairs has probability 0.10. Their means are 7, 8, 9, 9.5, 10.5, 11.5, 12, 13, 14, and 15.5, so the dotplot contains one point at each value.

Averaging those 10 sample means gives 11 hours, exactly the population mean. This establishes unbiasedness for the stated sampling process, but unbiasedness does not guarantee that every individual estimate is close. Only the six means from 9 through 13 lie within 2 hours of 11, producing probability 0.60 and contradicting the claimed 75% success rate.

Expected Points:

10

SERIES DO-NOT-REPEAT BANK UPDATE, APSTAT-FRQ5

· APSTAT-FRQ5-001: Unit 1 primary; Unit 3 secondary; double-blind, placebo-controlled completely randomized experiment on 90 community volunteers prone to virtual-reality nausea, assigning 45 to active acupressure wristbands and 45 to visually identical sham bands; binary 20-minute ride completion table with 34 versus 24 successes; identify population/sample/units/treatments/response, design random assignment, explain placebo and participant/evaluator blinding, calculate a 0.222 difference in completion proportions, and separate causal inference for volunteers from limited population generalization; targets assuming random assignment creates a representative sample and overlooking placebo/evaluator effects.

- APSTAT-FRQ5-002: Unit 2 primary; Unit 1 secondary; normal-model comparison of visual- and auditory-choice goalkeeper reaction times with means 480 and 360 ms, SDs 40 and 30 ms, and observed times 420 and 318 ms; calculate and compare $z = -1.50$ and -1.40 , obtain percentiles 0.0668 and 0.0808, calculate a central visual probability of 0.6827, derive the fastest-5% cutoff 414.2 ms, and question the model under right skew; targets treating a low time percentile as poor performance and applying normal tails despite incompatible shape.
- APSTAT-FRQ5-003: Unit 4 primary; Unit 1 secondary; independent two-sample t confidence interval comparing standardized safe-temperature duration for insulated lunch containers from Suppliers A and B, using independent SRSs with $n_A = 22$, $\bar{x}_A = 9.84$, $s_A = 0.62$ and $n_B = 20$, $\bar{x}_B = 9.31$, $s_B = 0.71$; check independence/10%/shape, use $df = 38$ and $t^* = 1.686$, calculate $SE \approx 0.2066$ and 90% interval (0.182, 0.878) hour, then distinguish evidence of a positive advantage from evidence of at least 0.25 hour; targets using a paired procedure and overclaiming a minimum effect from an interval.
- APSTAT-FRQ5-004: Unit 1 primary; Unit 2 secondary; finite population of five theater-backdrop repainting times 6, 8, 10, 13, and 18 hours; enumerate all 10 equally likely SRSs of size 2 and their unique sample means, construct and sketch the exact sampling distribution, compare population mean 11 with median 10, show $E(\bar{x}) = 11$ and unbiasedness, and calculate $P(|\bar{x} - 11| \leq 2) = 0.60$ to reject a 75% accuracy claim; targets confusing unbiasedness with high probability of a close estimate and counting ordered samples as distinct.